MASTER FUNCTIONAL SKILLS

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NCFE · LEVEL 1 · MATHEMATICS

NCFE Functional Skills Mathematics Level 1

October 2019 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-1-Oct-2019-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Examiner use only

Activity 2: Community nurse

2 (a) Asmita is a community nurse.

Each week she runs a drop-in clinic where people can have a check-up. People arrive at the clinic and wait until it is their turn to be seen.

At one clinic Asmita asks each person to estimate how many minutes they have had to wait.

Q2(a) Worked Solution:

The results are shown below.

See below
1

2	15	18	32	37	8	25	13
31	32	3	5	6	7	4	12
12	14	11	35	28	11	8	12

3 or more non-overlapping class intervals

Use this template to create a grouped frequency table for this data.

[4 marks]

(e.g. 0-9,10-19, 20-29, 30-39 – see table below)
Allow intervals of different sizes
Allow if additional (unnecessary) classes added
H28a
1
[4 marks]

past paper

Please turn over

Examiner use only

2 (b) At the next clinic, the time that each person arrives is recorded.

Asmita also writes down the times that each person's check-up begins and ends.

She uses this information to draw a graph.

Q2(b) Worked Solution:

See below

1

40

15

100

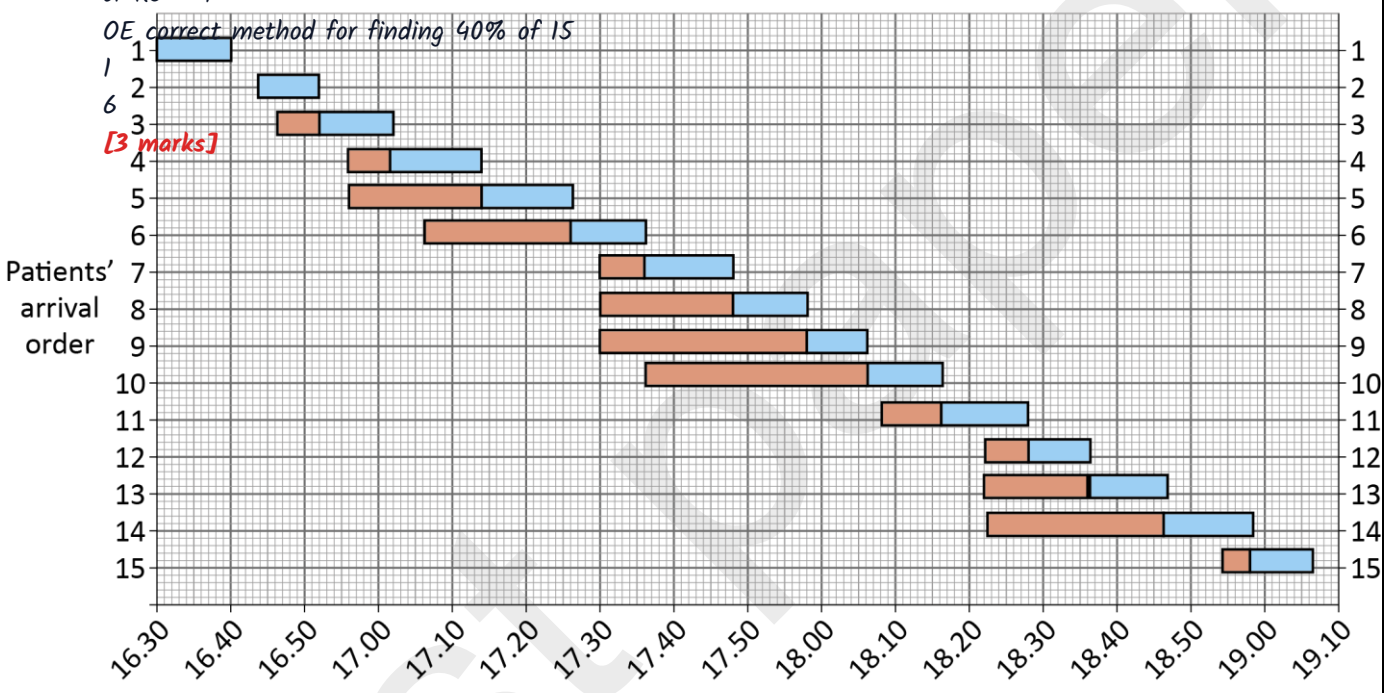
or 15 × 4

OE correct method for finding 40% of 15

1

6

[3 marks]



past paper

Examiner use only

Asmita says,

"40% of people either waited no time or waited less than 10 minutes before the start of their check-up."

Is Asmita correct?

Show your working.

[3 marks]

Your answer:

past paper

Please turn over

Examiner use only

2 (c) Asmita does home visits.

She drives from Skegness to Candlesby to visit one of her patients.

Asmita has a map with scale 1 cm : 2 km

Q2(c) Worked Solution:

The distance on the map between Skegness and Candlesby is 6.5 cm

13 and (£)3.64 in table

Asmita can claim 28p per kilometre for fuel.

Complete the travel expense form below for Asmita's **one-way** journey from Skegness to Candlesby.

[3 marks]

Award 3 marks if correct answer given

1

Your answer:

13 (km) in table

1

Journey	Distance (km)	Pence per km	Total (£)
Skegness to Candlesby		28	

their 13 × 28 or 364
or their 13 × 0.28 or 3.64

"Their" is from the value calculated in 1st mark

1

[3 marks]

past paper

Examiner use only

2 (d) Asmita parks her car on the road in Candlesby.

The car is facing north west.

When she has finished the visit, Asmita turns her car round ready to drive south east.

Q2(d) Worked Solution:

Through how many degrees has the car turned?

180 (°)

[1 mark]

[1 mark]

Your answer: °

2 (e) Asmita is concerned about how much fuel she is using.

The manufacturer says that the car has:

- a 52-litre fuel tank
- fuel consumption of 17 km per litre.

Q2(e) Worked Solution:

Asmita finds that she uses one full tank of fuel to drive 800 km

See below

1 Is Asmita's fuel consumption as good as the manufacturer claims?

Show how you decide.

[2 marks]

800 ÷ 52 or 15.38(...) or 15.4
or 52 × 17 or 884
1
15 or 15.38(...) or 15.4 or 884
and no
Award 2 marks if correct answer given

[2 marks]

Your answer:

Please turn over

past paper

Examiner use only

2 (f) Asmita puts 40.2 litres of fuel in her car.

The fuel costs 130.1 pence per litre.

She is charged £52.70 for the fuel.

Q2(f) Worked Solution:

By rounding **litres of fuel** and **cost per litre** to the nearest whole numbers, show how Asmita could check that she has been charged about the right amount.

[2 marks]

See below
1
40 × 130 (=5200)
1
52 or 5270 or (£) 52.70 seen
No FT on their 5200
8
Level 1 – P001253 (v1.1)
Activity 3:
French flag
[2 marks]

Your answer:

[Total marks: 15]

past paper

Examiner use only

Activity 3: French flag

3 (a) Tomas is an apprentice gardener.
He is asked to make a flowerbed to look like the French flag.

The flowerbed is a rectangle 225 cm long and 150 cm wide.

Tomas is going to put paving round the perimeter of the flowerbed.

Calculate the perimeter of the flowerbed in metres.

[2 marks]

Q3(a) Worked Solution:

7.5 (m)
Award 2 marks if correct answer given
1
225 + 225 + 150 + 150 or 750
or 2.25 + 2.25 + 1.5 + 1.5 or 7.5
1
7.5 (m)
[2 marks]

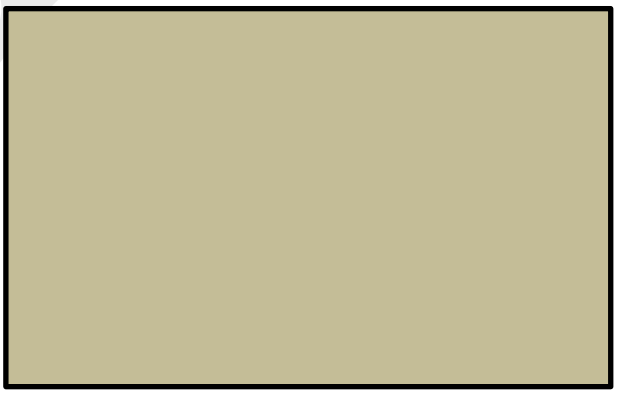
Your answer: **m**

3 (b) Draw in all the lines of symmetry on the flowerbed.

[1 mark]

Q3(b) Worked Solution:

Mark intention. Both lines of symmetry required.
[1 mark]

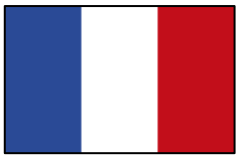


past paper

Please turn over

Examiner use only

3 (c) Tomas will use red, white and blue flowers to make the flowerbed look like the French flag.



- The flowerbed is a 225 cm by 150 cm rectangle.
- The 3 stripes need to be of equal width.
- Tomas needs one plant for every 12 cm^2 of flowerbed.

How many red plants will Tomas need?

[4 marks]

Your answer:

Q3(c) Worked Solution:
937 or 938 (plants)
Award 4 marks if correct answer given
1

225 × 150 or 75 × 150
or 33750 or 11250
Method for finding area of flag or stripe
1
11250
1
their 11250 ÷ 12 or 937.5
[4 marks]

past paper

Examiner use only

3 (d) Tomas's manager is ordering plants for a different project.

The table shows the costs per tray for different colours.

Flower colour	Cost per tray
Blue	£42.98
Red	£31.56
White	£33.98
Yellow	£42.99
Pink	£35.75
Purple	£28.50

Q3(d) Worked Solution:

Calculate the mean cost per tray.

[2 marks]

(£)35.96
Award ~~2 marks if correct answer given~~
1
(42.98 + 31.56 + 33.98 + 42.99 + 35.75 + 28.50) ÷ 6 or 215.76 ÷ 6
H29a
1
(£)35.96
H29a
[2 marks]

Your answer: **£**

past paper

Please turn over

Examiner use only

3 (e) The order is for one tray of each colour.

There are 310 plants in each tray.

Tomas says, “The mean cost per plant is over 10 pence.”

Is Tomas correct?

Explain your answer.

[2 marks]

Q3(e) Worked Solution:

11.6p or £0.116 and yes
Award 2 marks if correct answer given
1
their 35.96 ÷ 310 or 3596 ÷ 310
or their 215.76 ÷ (310 × 6)
or 21576 ÷ (310 × 6)
or (£) 0.116 or 11.6 (p)
FT from “Their” final value for Q3d
H29a
1
[2 marks]

Your answer:

past paper

Examiner use only

3 (f) Each plant tray is cuboid and measures 30 cm by 50 cm by 4 cm

How many litres of compost would be required to fill one tray?

Use the conversion 1 cm^3 = 1 ml

[2 marks]

Q3(f) Worked Solution:
6 (litres)
Award 3 marks if correct answer given
1
30 × 50 × 4 or 6000 (cm3)
1
6 (litres)
9
Level 1 – P001253 (v1.1)
[2 marks]

Your answer:

past paper

Please turn over

Examiner use only

3 (g) Tomas needs to mix grit and compost in the ratio 1 : 4

He mixes $4\frac{1}{5}$ litres of grit with $16\frac{4}{5}$ litres of compost.

Are these numbers in the correct ratio?

Q3(g) Worked Solution:

Show how you decide.

[2 marks]

4.2 x 4 = 16.8 and yes OR 21:84 = 1:4
and yes
Award 2 marks if correct answer given
1
4.2 and 16.8
OR 21
84
and
5
5
[2 marks]

Your answer:

[Total marks: 15]

past paper

Examiner use only

Activity 4: World hunger

4 (a) The world population is about 7726.5 million.

About one in nine people in the world experience extreme hunger.

Use this information to estimate how many millions of people in the world experience extreme hunger.

Give your answer to the nearest million.

[2 marks]

Q4(a) Worked Solution:

See below

1

7726.5 ÷ 9 or 7726 500 000 ÷ 9

1

859 (million)

[2 marks]

Your answer: **million**

past paper

Please turn over

Examiner use only

4 (b) When people are starving, aid organisations have to decide quickly who to help first.

One method they use is to measure, in mm, the distance round each person's mid upper arm.

These are the mid upper arm measurements of 20 people.

Q4(b) Worked Solution:
39
H29b

170	172	158	175	156
154	162	151	172	186
158	173	190	174	169
178	175	157	188	189

[1 mark]

What is the range of the measurements?

[1 mark]

Your answer:

4 (c) What is the probability that one of these people chosen at random has a mid upper arm measurement of less than 160 mm?

Give your answer as a decimal.

[2 marks]

Q4(c) ~~Worked Solution:~~
0.3
Award 2 marks if correct answer given
1
6
20
or 3
10
H31
1
0.3
[2 marks]

Your answer:

past paper

Examiner use only

4 (d) Another method aid organisations use is to calculate Body Mass Index (BMI).

To find the BMI:

Step 1: calculate height × height

Step 2: divide weight by your answer to Step 1

Q4(d) Worked Solution:

25 Height must be in metres and weight in kilograms.

Calculate the BMI for a person who has:

- weight 64 kg
- height 1.60 m

Award 3 marks if correct answer given **[3 marks]**

1

1.60 × 1.60 or 2.56

1

64 ÷ their 2.56

FT their 2.56 from first mark

1

25

[3 marks]

Your answer:

past paper

Please turn over

Examiner use only

4 (e) An aid charity finds that:

20% of the people in village A are malnourished.
15% of the people in village B are malnourished.

Village A has a population of 300
Village B has a population of 440

Which village has the higher number of people who are malnourished?

Show how you decide.

[2 marks]

Q4(e) Worked Solution:

See below
10
Level 1 – P001253 (v1.1)
1
0.2 × 300 or (300 ÷ 10) × 2 or 60
or 0.15 × 440 or 440
440
1
10
10
[2 marks]

Your answer:

4 (f) A newspaper reports that the UK wastes one million tons of food each year which could be used to feed hungry people.

Write one million in figures.

[1 mark]

Q4(f) Worked Solution:

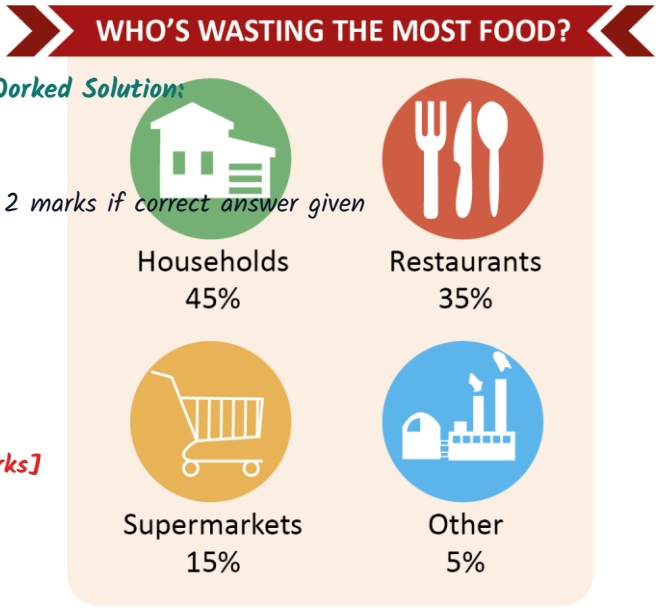
1 000 000
CAO Ignore commas.
Do not allow full stops/decimal points.
[1 mark]

Your answer:

past paper

Examiner use only

4 (g) This information appeared on a website about food wastage.



Q4(g) Worked Solution:
3
20
Award 2 marks if correct answer given
1
15
100
1
3
20
[2 marks]

What fraction of food wastage came from supermarkets?

Give your answer in its lowest terms.

[2 marks]

Your answer:

past paper

Please turn over

Examiner use only

4 (h) Use the information about food wastage given in **4 (g)** to complete the pie chart below.

[2 marks]

Q4(h) Worked Solution:

See below

1

All sectors correct size but labels (or key for onscreen version) missing or 2 sectors correct and correctly labelled.

Supermarkets 3 segments OR 54 (degrees)

Restaurants 7 segments OR 126 (degrees)

[2 marks]

Who's wasting the most food?



[Total marks: 15]

This is the end of the external assessment.

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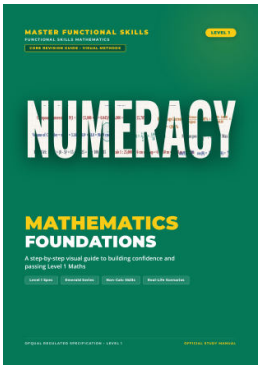
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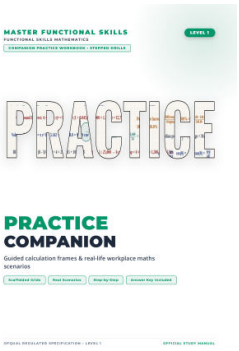


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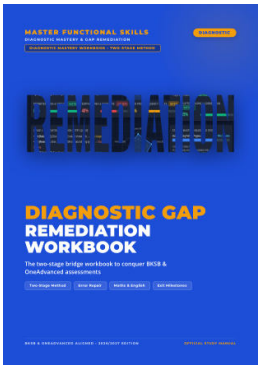


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